

The use of velocity-based training in strength and power training

- A systematic review

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Abstract

Background: The intensity or load of a strength training exercise is commonly considered to be the most important factor contributing to muscular strength and power. Traditionally in strength training, intensity is defined as the percentage of the maximum weight that can be lifted once i.e. 1 repetition maximum. For power development exercises, the velocity can be used to measure the intensity. A linear position transducer is able to measure kinetic and kinematic variables. Velocity-based training refers to the usage of a linear position transducer to track movement velocity of an exercise and thus, using velocity, rather than load, as a measurement of intensity.

Purpose: The purpose of this systematic review was to provide an analysis of the existing velocity-based training research utilizing a linear position transducer. The study also aimed to investigate the validity and reliability of different commercial linear position transducers for kinetic and kinematic measurements.

Method: A systematic review was conducted from 19 studies on velocity-based training that met the selection criteria and underwent a quality assessment.

Results: It was possible to predict the 1 repetition maximum using velocity and the minimal velocity threshold was stable across different relative intensities. Performing squats at either maximal velocity, or stopping at a velocity loss of <40% could significantly improve 1 repetition maximum, increase mean velocity during a set of squats as well as vertical jump performance. Two linear position transducer were found to have excellent validity and reliability for both kinetic and kinematic measurements.

Conclusion: Velocity-based training was beneficial for enhancing neuromuscular adaptations and could be used to predict the 1 repetition maximum. When using of a linear position transducer for power development, it is suggested that it is valid and reliable for both kinetic and kinematic measurements.

Keywords: force-velocity, linear position transducer, power, velocity, velocity-based training

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Introduction

The outcome of strength training and subsequent adaptations are influenced by a variety of factors. The level of the athlete, sex, muscle activation type, exercise selection and loading strategies are some of the many factors that contribute and effect the adaptations to training (Stone, Cormie & Lamont 2016, p. 230). Evidence suggest that the mode (e.g. variation, progression, specificity) and method (e.g. free-weight, machines, body-weight) in which training is performed by the athlete can make a profound difference in the performance outcome (Stone et al. 2002). Muscular strength is defined as the ability of the neuromuscular system to produce force against an external resistance (Bompa & Haff 2009, p. 261). The importance of force production can be ascertained from Newton's second law of motion: $F = ma$ in which the acceleration of a body mass or an external object relies on the musculatures ability to generate force. Simply put, acceleration of an object is only doable when enough force is applied to it. This makes muscle force a major component when attempting to generate effective and efficient movement of the body or an external object. Furthermore, strength and force production is accompanied by a rate of force development, a power output and velocity of movement. (Stone, Cormie & Lamont 2016, p. 230).

There are many different methods of resistance training that can be utilized to develop an athlete's strength and power characteristics, e.g. weightlifting, plyometric training, eccentric training and ballistic training (Suchomel et al. 2018). Additionally, variable resistance training using chains are commonly used tools to better accommodate resistance to the biomechanical characteristics of a particular exercise. Using variable resistance is an attempt to alter the load during certain phases of the range of motion (Godwin, Fernandes & Twist 2018; McMaster, Cronin & McGuigan 2009). Other training techniques for strength and power development are cluster-set training (allowing intra-set rest periods to maximize intensity and minimize fatigue), complex training (heavy loading followed by a lighter plyometric-type movement) and contrast training (alternating heavy loads with light loads) (French 2017, pp. 182-197).

Velocity-based training is a method for strength and power development. In velocity-based training one uses a piece of technology, called linear position transducer, to track the movement velocity of an exercise and thus, uses the velocity as a measurement of intensity. The reason for the interest of velocity-based training may be attributed to its ability to report real-

time feedback on a repetition-to-repetition basis, which allows athletes and coaches to dictate the speed at which an exercise should be performed at (Jovanović & Flanagan 2014).

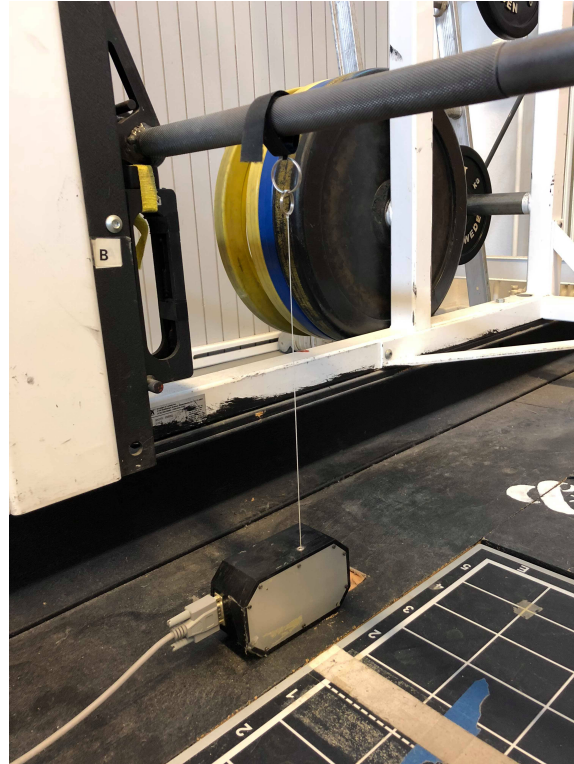


Figure 1. A linear position transducer attached to a barbell

Measuring intensity

In resistance training, the intensity of an exercise is commonly considered to be the most important factor contributing to muscular strength and athletic performance (Kraemer, Fleck & Deschenes 1988). This has been highlighted by Mangine et al. (2016) who found that a higher intensity was favoured over high volume in order to increase force and rate of force development. A problem in strength and conditioning is the ability to objectively quantify and monitor training intensities (González-Badillo & Sánchez-Medina 2010).

In resistance exercise training, intensity is often defined as the percentage of maximal strength (%1 repetition maximum [%1RM]) for a particular exercise. This definition is sometimes also referred to as relative intensity, since it is based on a person's strength capacity (Fry 2004). Assessment of the 1RM has several drawbacks, for example, athletes are at risk for injuries when using maximal weights. The 1RM test is also fairly time consuming and, further, the result of the 1RM can change from day-to-day through fatigue, biological varia-

bility and life-style factors (González-Badillo & Sánchez-Medina 2010). Another way to quantify intensity is through the use of RM loads. These are based on the most weight an individual can lift for a prescribe number of repetitions, e.g. 8 RM (Baechle, Earle & Wathen 2008, p. 395). Intensity can also be defined as percentage of effort and relies on the individual's perception of their own level of exertion to determine intensity (i.e. RPE-scale) (Day et al. 2004).

A different method to measure intensity is through the use of velocity using a linear position transducer (Conceição et al. 2016; González-Badillo & Sánchez-Medina 2010; Bazuelo-Ruiz et al. 2015). Velocity specificity in resistance training suggests that adaptations following training are maximized at or near the velocity of movement used during training (Pereira & Gomes 2003; Murray et al. 2007; Negra et al. 2016). There also exists a theory in which the intention to move at a high velocity is more important than the actual velocity (Pereira & Gomes 2003). While it takes several seconds to complete a single repetition in the majority of weight-room exercises, rarely in sport is time sufficient enough to reach maximal force, therefore, velocity specificity is suggested to be a central consideration when training to develop muscular power for sport performance (McBride et al. 2002).

How does a linear position transducer work?

Linear position transducers are used to measure strength and power in resistance training exercises. The device works by attaching a cord to either the equipment used (e.g. a barbell), or directly to the subject performing the exercise (e.g. a vertical jump). As the person or object moves, the displacement is measured and recorded with a computer system (Harris et al. 2010). Data on displacement, velocity of movement and acceleration is collected. The velocity is calculated through the differentiation of displacement and time ($\text{velocity} = \text{displacement} / \text{time}$). Furthermore, double differentiation of the displacement makes is possible to calculate acceleration ($\text{acceleration} = \text{velocity} / \text{time}$) which allows for the determination of force ($\text{force} = \text{mass} / \text{acceleration} + \text{acceleration due to gravity}$). Additionally, the linear position transducer uses inverse dynamics to calculate power output solely from displacement data ($\text{Power} = \text{velocity} \times \text{force}$) (Garnacho- Castaño, López-Lastra & Maté-Muñoz, 2015; Newton, Cormie & Cardinale 2011, p. 264).

The linear position transducer typically takes the form of a central processing unit with a retractable measuring cable and is connected to a computer with a software that is generally

available or customized for a specific use (Jovanović & Flanagan 2014). It usually consists of four main parts: measuring cable, spool, spring and a rotational sensor. A linear position transducer samples and records data during the measurement period of interest. Sampling frequency or sampling rate is the number of data points collected every second and are measured in hertz (Hz). It has been suggested that a minimum sampling rate of 200 Hz is used, but 500-1000 Hz is desirable. With a slow sampling rate (<200 Hz) during very fast movements, the chances that peaks do not get recorded. Furthermore, the smoothing and/or filtering is of great essence for the validity of the linear position transducer when measuring force outputs due to the fact that errors occur when first and second derivative are calculated (Harris et al. 2010).

When using a linear position transducer, there are different metrics that need to be acknowledged. Mean concentric velocity refers to the average speed during the entire concentric phase of the exercise. Peak concentric velocity is simply the peak speed during the concentric phase. Whereas mean concentric velocity is more suited for typical strength base exercises (e.g. deadlift, squat and bench press), peak concentric velocity is used for power-based exercises (e.g. Olympic lifts). Lastly, mean propulsive velocity is the portion of the concentric phase where the acceleration is greater than the acceleration due to gravity ($-9,81 \text{ m} \cdot \text{s}^{-2}$). When using light and medium loads, there exist a deceleration phase at the end of the lift which may influence the results. Measuring the mean propulsive velocity will provide the individual with a better reflection of its neuromuscular ability as suggested by Sanchez-Medina, Perez & Gonzalez-Badillo (2010).

The relationship between force and velocity

Mechanical power can be defined as the product of force multiplied by velocity ($P = F \times V$). The force-velocity relationship is the observation that muscle force and contraction velocity are inversely related. Because of this relationship between force and velocity, maximal power output will occur at compromised levels of maximal force and velocity (Haff & Nimphius 2012) (figure 1).

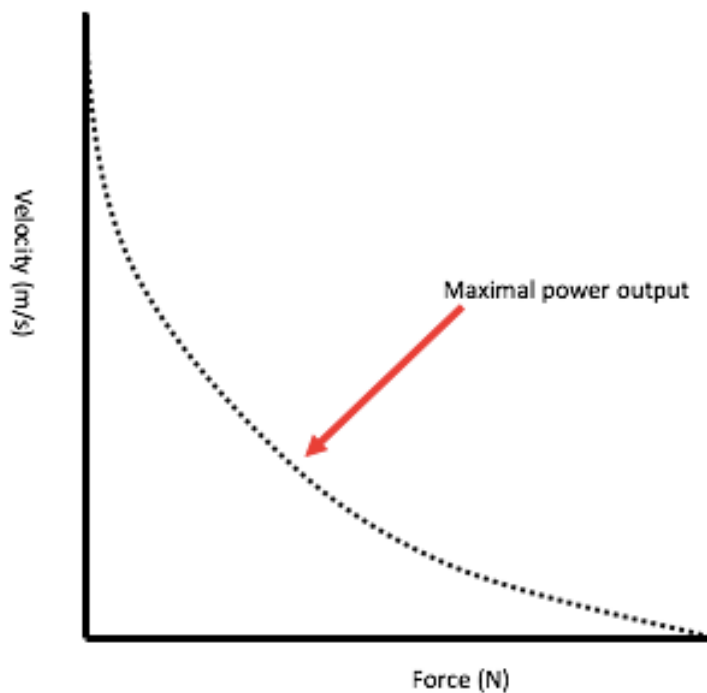


Figure 2. Redrawn from Haff & Nimphius (2012). The force-velocity relationship.

Individuals may experience different balances between force and velocity, even if they may have the same power output. With a linear position transducer, it is possible to determine whether or not an individual has a velocity-deficit (power outputs comes mainly from force production) or force-deficit (power outputs comes mainly from velocity production) (Jovanović & Flanagan 2014).

Samozino et al. (2012, 2014) suggest that there exists an optimal balance between force and velocity for each individual that will maximize power output, this is also known as force-velocity profile. Through force-velocity profiling, a coach can identify whether an athlete is force- or velocity-deficient during a given movement (e.g. vertical jump), independent of their power capability, and thereafter, individualize the training.

In order to change the shape of the force-velocity curve one has three theoretical options: achieve training induced increases of either velocity, force or increases of both parameters (Haff & Nimphius 2012). In a study by Jiménez-Reyes et al. (2017) 84 subjects were profiled as either velocity-deficient, force-deficient, or well-balanced. From there, subjects were broken into five different sub-groups of force-deficit group, velocity-deficit group, a well-

balanced group, a non-optimized group and a control group. The aim of this study was to see if an individualized training program based on the individual force-velocity profile would decrease the subjects force-velocity imbalanced (FV_{imb}) and in turn improve vertical jump performance after following a training program for 9-weeks. All subjects in the three optimized training groups increased their jumping performance, while the results were very unclear and variable in the non-optimized group. The authors conclude that an optimized and individualized training program specifically addressing the force-velocity imbalances is more efficient at improving jumping performance than a traditional resistance training program common to all subjects regardless of force-velocity imbalances. This method could be considered a useful variable for prescribing resistance training programs to individuals. McMaster (2016) investigated the force-velocity profile among 20 semi-professional rugby players and found that the forwards were more force dominant, whereas the backs were more velocity orientated when tested in bench push.

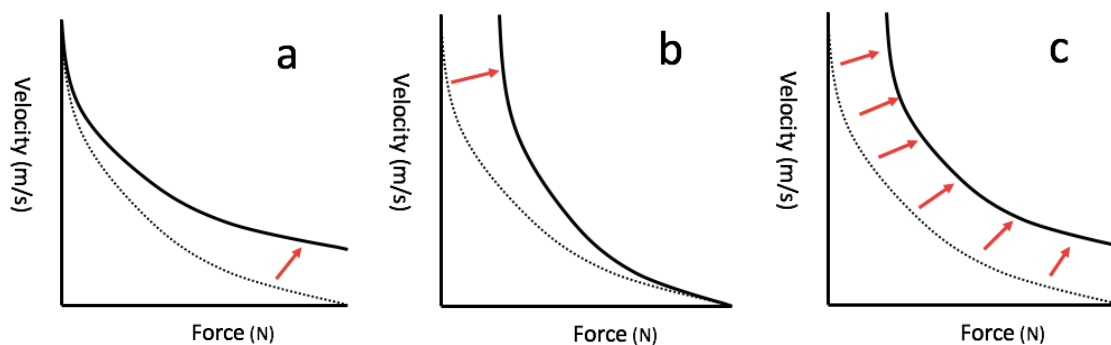


Figure 3. Redrawn from Haff & Nimphius (2012). a) increases of force on the force-velocity curve. b) increases of velocity on the force-velocity curve. c) increase of both force and velocity.

From the force-velocity profile it is possible to determine the optimal load which has been defined as the load at which the individual produces the highest amount of power (Newton, 2011, p. 91). To most effectively increase power output, it is thought that athletes should train with a resistance that results in the highest power output possible (Kawamori & Haff, 2004). In theory, optimal loads provide an ideal combination of force and velocity magnitudes that produce high power outputs. However, the optimal load seems to be very exercise specific

for both upper and lower body exercises (Soriano et al. 2016; Soriano et al. 2015). Haff and Nimphius (2012) suggest the prescription of a range of loads near the “optimal load” in order to develop both low and high force/power characteristics during different exercises.

When reaching failure, the movement velocity will be zero. Monitoring the velocity with a linear position transducer makes it possible to track how close to failure the subject is. As the 1 repetition maximum (1RM) refers to the maximum weight that can be lifted once, the velocity will be close to zero at that point. Given this information, using sub-maximal loads and calculating the slope of velocity, it is possible to predict the 1RM. This method has, as mentioned earlier, several benefits compared to the traditional ways of assessing the 1RM (González-Badillo & Sánchez-Medina 2010). The investigations that have examined this, have done so by utilizing a variety of different protocols and there is a need to further examine this.

Based on the same assumption that when reaching failure, the velocity is going to be zero, the minimal velocity threshold can be applied. The minimal velocity threshold is the mean velocity produced on the last successful repetition of a set to failure which is usually assessed though a traditional 1RM or with with submaximal loads. When this is known, it can be used to predict how close to failure an individual is by monitoring the velocity of each repetition. This can be beneficial for both 1RM assessments and for general training sessions to avoid failure (Jovanović & Flanagan 2014).

A number of studies has suggested that reaching failure may not be necessary to improve the magnitude of strength gains, especially when aiming for neuromuscular adaptations rather than strength gains through muscle size gains (Drinkwater et al. 2007; Folland et al. 2002; Izquierdo et al. 2006). Furthermore, fatigue could be defined as the transient decline in muscle force generating capacity (Enoka & Duchateau 2008). Fatigue will not only affect force production, but also decrease the shortening- and relaxation- speed of the muscle (Allen, Lamb & Westerblad 2008). As both force production and velocity is negatively affected by fatigue, power output will decrease.

Sánchez-Medina and González-Badillo (2011) have used the term “level of effort” to define the actual number of repetitions performed in a set in relation to the maximum number that can be completed (i.e. proximity to muscle failure). Several studies have investigated the effects of different levels of effort protocols (Morán-Navarro et al. 2017; González et al. 2016;

Sánchez-Medina & González-Badillo 2011) and found that the protocols leading to failure (higher level of effort) have a higher biochemical stress, greater fatigue and a slower rate of neuromuscular recovery compared to the protocol not leading to failure (lower level of effort). As athletes may participate in more than one training session a day, being able to recover faster while gaining the same results is of great essence for sport performance.

When measuring the velocity of each repetition, it is possible to objectively quantify the level of effort at any given moment, hence controlling the extend of fatigue. Knowing this, is is possible to reasonably estimate the neuromuscular and metabolic stress induced by the exercise (Jovanović & Flanagan 2014).

Gold standard

In classical mechanics, kinematic is defined as the description of movement independent of forces, whereas kinetics refers to the forces that cause movement (Winter 2009, p. 10). Force plates has been referred to as the “gold standard” for the direct acquisition of kinetic data (Nigg & Herzog 2007, p. 324). This modality relies on the impulse–momentum relationship to determine velocity and allows for power to be calculated through a forward dynamics approach. When a subject push against a force plate during dynamic or static movement, data can be obtained. If the equipment is fixed (meaning that no movement can occur) one can measure isometric peak force and rate of force development. Calculating power from a vertical ground reaction force via a force plate is commonly used in strength and power research to compare various types of body weight jumps or to monitor vertical jump performance following an intervention (Newton, Cormie & Cardinale 2011, 264).

Cormie et al. (2007) tested the validity of power measurement techniques utilizing various kinematic and kinetic devices during the jump squat, squat and power clean in ten division I male athletes. During the execution of each lift, six different data collection systems were utilized. They concluded that methodologies relying solely on kinematic data consistently both elevated and underestimated power output across various loads in the exercises. Using only the force plate was shown to underestimate the velocity and power output, especially during movements involving the bar traveling independently of the body, such as in the power clean. Preferably, the utilization of both a force plate for kinetic measurement and a linear position transducer for the measurement of velocity, this notion is also supported by Nigg & Herzog (2007, p. 347). It is suggested that the use of a force plate and two linear position

transducers to assess both vertical and horizontal displacement during multi-dimensional movements (especially during weightlifting movements) (Cormie et al. 2007). When using only one linear position transducer with or without a force plate, during weightlifting exercises, up to 10% of the work can be done horizontally which could result in an inflated velocity and power measurement (Garhammer 1993). Hansen and colleagues (2011) compared the reliability between using a linear position transducer, a force plate and the combination of the two when performing three jump squats on two separate occasions. The results showed that the combined method and the force plate had the greatest reliability (ICC 0.95-0.94) compared to the lower reliability found in the linear position transducer (ICC 0.87) for peak power. When measuring peak velocity, the force plate was superior (ICC= 0.93) over the linear position transducer (ICC= 0.89). However, the correlation between all three methods was found to be high for peak power ($r = 0.81-0.95$) and a bit lower for peak velocity ($r = 0.67-0.76$). In table 1, Nimphius (2017, p. 20) presents several advantages and disadvantages with different measuring instruments for the measurement of mechanical power.

Table 1. Redrawn from Nimphius (2017, p. 26). Advantages and disadvantages of different power measurements. LPT= linear position transducer, FP= force plate.

Equipment	Advantages	Disadvantages
2-4 LPTs and FP	Direct measure of force and displacement (to calculate velocity). Includes measurements of horizontal displacement (or velocity) during movements that are not fully linear (vertical or horizontal only).	Most expensive setup and requires designated space because of equipment requirements.
1 LPT and FP	Direct measure of both force and displacement (to calculate velocity). Combination is valid assessment of power based on the bar velocity during ballistic movements.	Assumption that movements are fully linear (e.g., not ideal for Olympic variations) and bar velocity is representative of centre of mass velocity.
FP	Highly reliable and valid for both weightlifting (from hang positions) and ballistic movements. Valid assessment of the power based on the centre of mass velocity.	Mathematical manipulation to calculate power that could result in error associated with underestimation of power.
1 LPT or 2-D kinematic analysis	Relatively inexpensive. Reliable measure of velocity if movement is primarily linear (vertical or horizontal only).	Mathematical manipulation can cause magnification of error in calculation and often overestimation of power.
Accelerometer	Relatively inexpensive. Reliable measure of jump height during CMJ.	Reliability and bias of velocity and power measures increases minimal differences required when assessing changes in power.

Description of the problem area

The use of movement velocity as a measurement of intensity has not been widely investigated in the research literature (González-Badillo & Sánchez-Medina 2010), and most of the existing research has been obtained with isokinetic exercises where the muscle contracts at a constant and consistent rate of speed (Kraemer & Ratamess 2004). As the number of commercial linear position transducer increases, and the use of so called velocity-based training regimes in sport performance settings, so does the research and the need to overview this field. To the authors knowledge, there is only one study that has reviewed the benefits of monitoring movement velocity in strength training exercises (Jovanović & Flanagan 2014). It could be argued, however, that the proposed benefits presented by Jovanović and Flanagan (2014) needs to be analysed more deeply before any conclusion can be drawn regarding their

value. Furthermore, no review has yet looked into the different linear position transducer available. With the current increased in popularity of linear position transducer in sports and exercise, it is vital to more thoroughly determine the validity and reliability of this method.

Purpose

The aim of this study was to use a systematic review to provide a critical, constructive analysis of the existing research utilizing a linear position transducer through summary and analysis of previously published literature and data. Furthermore, to investigate the validity and reliability of different linear position transducers for kinetic and kinematic measurements. Information gathered from this paper will be useful to strength and conditioning coaches (and athletes) when utilizing a linear position transducer. Hence, this review tried to answer the following questions:

- Can repetition velocity really predict the 1 repetition maximum?
- What is known of the minimal velocity threshold?
- Validity and reliability of different linear position transducers?
- What are the neuromuscular effects of velocity-based training?

Method

Data collection

This systematic review is based on previously published literature. Data was primarily collected from the databases PubMed, SportDiscus and ResearchGate. The main keywords used was a combination of: *linear position transducer*, *velocity based training*, *power development*, *movement velocity*, *force-velocity*, *1 repetition maximum*. In addition, hand searches of books as well as the reference lists of articles already obtained were performed. Lastly, relevant research that was not found in any of the aforementioned databases was obtained via direct communication with the author(s) (see Appendix 1).

Inclusion and exclusion criteria

For inclusion in this review, all studies had to be relevant to the aim of the study and published in English. Further, studies must have examined healthy human subjects. Only original articles that had been peer-reviewed were included in this review. Both cross-sectional and experimental study designs were allowed in this study. No date restrictions were used to maximise record retrieval. One article could be used to answer several questions. The results in the studies had to be described at least to some detail (e.g. absolute values, relative values, graphs and/or figures). Finally, all subjects had to have previous experience in resistance training of at least 6 months. If the resistance training experience was not specified, but the subjects were athletes, it was considered valid enough.

Articles were excluded if the subjects were living under catabolic conditions during the time of the study. Furthermore, investigations that did not use a linear position transducer to estimate velocity were excluded.

A list of all the included articles can be found in Appendix 2.

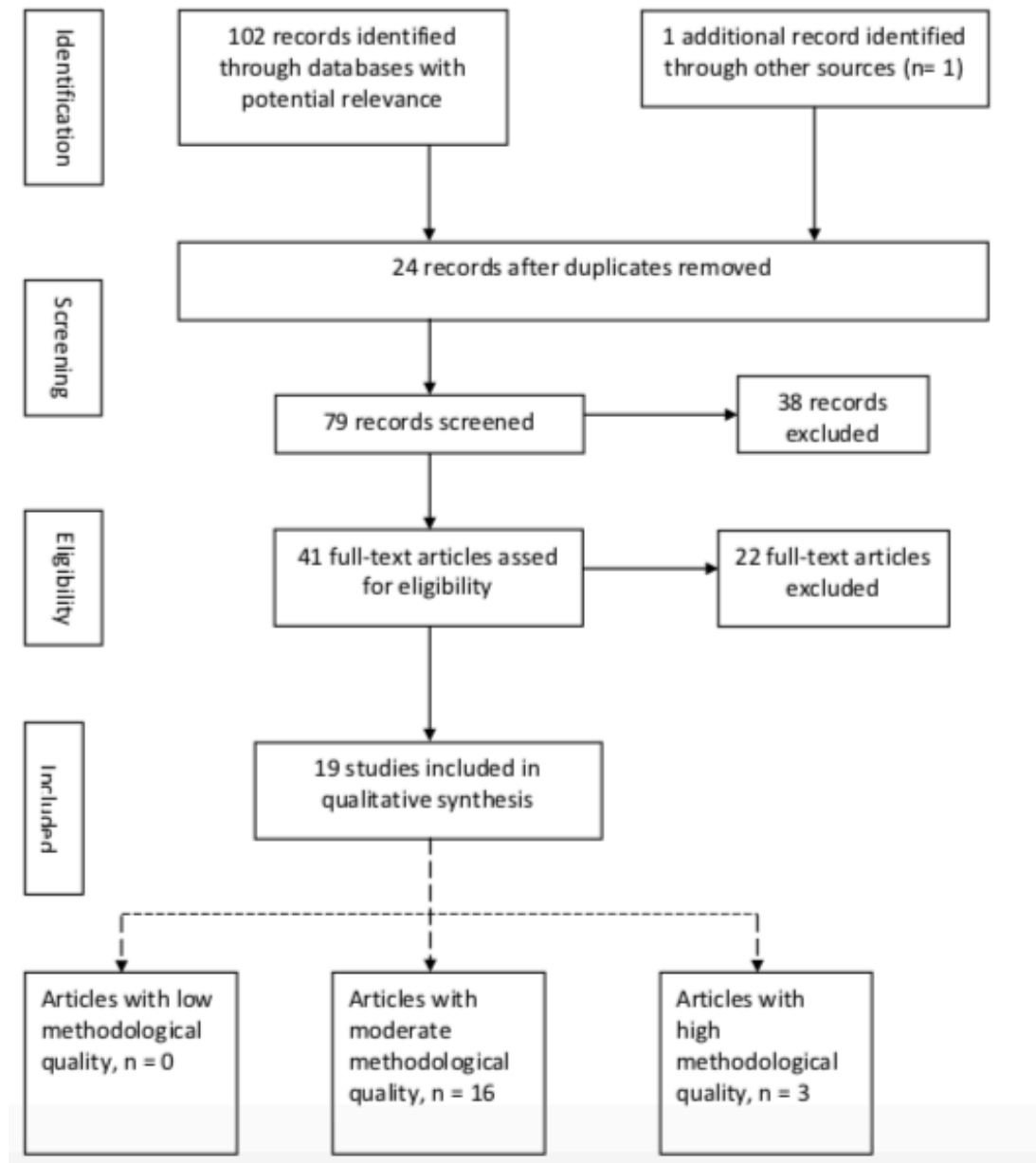


Figure 4. Flow of diagram of the studies that underwent this review process.

Quality assessment

To grade the quality of the included studies, the methodological quality of all studies were assessed with the Oxford Centre of Evidence Based Medicine levels of Evidence, 2001. This is an arranged ranking system used in evidence-based practices to describe the strength of the results measured in a research study. The study's methodological quality was quantified as high (evidence grade A), moderate (evidence grade B) or low (evidence grade C and D).

Table 2. Study design and corresponding level of evidence. Adapted from: Oxford Centre for Evidence-Based Medicine, 2001. SR = systematic review; RCT = randomized controlled trial.

Evidence Grade	Level of Evidence	Type of Evidence
A	1a	SR of RCT's
	1b	Individual RCT
	1c	All or None
B	2a	SR of Cohort Studies
	2b	Individual Cohort Study
	2c	Outcomes Research
	3a	SR of Case-Control Studies
	3b	Individual Case-Control Study
		Case Series (and poor quality case-control and cohort studies)
C	4	Expert opinion without critical appraisal or based on physiology or "first" principles
D	5	

Validity and reliability

Validation using Pearson's product moment of correlation coefficient (r), coefficient of determination (r^2) and adjusted coefficient of determination (r^2_{adj}) was considered poor for values less than 0.5, good for values between 0.5-0.75 and excellent for values above 0.75 (Almeida et al. 2017). For reliability, interclass correlation coefficient (ICC) was considered poor for values less than 0.69, fair for values between 0.7-0.79, good for values between 0.8-0.89 and excellent for values above 0.9 (Shrout 1998).

Results

Predicting the 1 repetition maximum

The results presented in table 3 suggest a load ranging from 20-95% of the previously known 1RM when attempting to predict the 1RM with the use of velocity. Banyard, Kazunori & Nosaka (2017) investigated the prediction of 1RM using loads up to 60%, 80% and 90% and found the 90% load to be the most reliable ($r = 0.93$, $r^2 = 86.49$). This gets supported by other studies (in the results) that have used loads up to 90% predict the 1RM. Conceição et al. (2016) was the only study that used a minimum of 6 loads. This may be due to the fact that they started the test with loads 20-30% of their previously known 1RM and then progressively loaded up until subject's reached failure. One advantage with this method is that more data

points are obtained during the assessment which could facilitate the 1RM prediction. A disadvantage may be the time demanding aspect of the procedure, and the possible risk of injury when reaching failure. However, the most common loads in the other studies were between 3-6. All studies had a minimum of 3-minutes rest between sets and only Conceição et al. (2016) had up to 6 minutes.

The correlation was found to be excellent ($r = 0.78-0.99$) for four out of five studies. The study by Conceição et al. (2016) presented their results in a different form (R^2 adjusted), but was considered to be a high value by the authors. The reliability was only presented by two studies (Banyard, Kazunori & Nosaka 2017; Ruf, Chéry & Taylor 2018). Loads up to 90% of 1RM had excellent reliability, whereas load up to 80% had good reliability and loads up to 60% only fair.

The first studies that tried to predict the 1RM using velocity, did so using a Smith machine which is a barbell that is fixed within steel rails allowing for only vertical movement and is usually counter-balanced. However, more recent published studies have investigated this relationship using free-weights and one study compared these two methods against each other. It is discussed by Banyard, Kazunori & Nosaka (2017) that when using free weight, the contribution of the stretch-shortening cycle may increase the power output, and thereby influence the results. However, there seems to be no difference in correlation between free weight ($r = 0.9772$, $r^2 = 95.51$) and Smith machine ($r = 0.9847$, $r^2 = 96.97$) exercises when predicting the 1RM in the bench press (Loturco et al. 2017). Similar results were found in the free deadlift ($r = 0.986$, $r^2 = 97.21$) and the back squat when performing up to 90% 1RM ($r = 0.93$, $r^2 = 86.49$).

To summarize, the results suggest that there exists a strong correlation between load and velocity and that it is possible to predict the 1RM using velocity as a measurement of intensity. However, there exist some controversy when using free-weights instead of a Smith machine.

Table 3. The correlation between velocity and load to predict the 1 repetition maximum. Significant correlation: MV= mean velocity, MPV= mean propulsive velocity, PV= peak velocity, r = correlation of coefficient, r^2 = correlation of determination, r^2_{adj} = adjusted correlation of determination, ICC= intraclass correlation. * $P= <0.05$, ** $P= <0.001$.

Study	Exercise	Subjects	Equipment	Protocol	V variables	r	r^2	r^2_{adj}	ICC
Loturco et al. 2017	Bench press	36 males	Smith machine	6 loads, 40-90%.		0.98	97		
				1-4 reps at each load. 3 min rest	MPV	0.98	96.6		
Loturco et al. 2016	Half squat	64 males	Smith machine	4 loads, 80-95%, 1-2 reps. 3 min rest		0.98	97		
					MPV				
Conceição et al. 2016	Leg press 115°	15 males	45° Leg press machine		PV		0.96*		
					MPV		0.96*		
				6-8 loads. 1-4 reps. 3-6 min rest	PV		0.94*		
					MPV		0.95*		
					PV		0.97*		
Banyard, Kazunori & Nosaka 2017	Back Squat	17 males	Free weight		MPV		0.96*		
				3-4 loads, 20-90%, 1-5 reps. 3 min rest	PV		0.94*		
					MPV		0.95*		
					PV		0.97*		
					MPV		0.96*		
Ruf, Chéry & Taylor 2018	Deadlift	11 males	Free weight	3-5 loads, 20-90%, 1-3 reps. 3-5 min rest		0.99	97		0.95-0.997
					MV				

Minimal velocity threshold

The minimal velocity threshold for the half squat seems to be between 0.30 to 0.33 m/s. Izquierdo et al. (2006) tested the minimal velocity threshold across 4 different loads leading up to failure, and found the minimal velocity threshold to be stable (0.31-0.31 m/s) independently of the relative load.

The minimal velocity threshold for the bench press was also stable across 4 different loads (60-75%), ranging from 0.17-0.18 m/s (Izquierdo et al. 2006). González-Badillo et al. (2017) examined the minimal velocity threshold with eight different loads (50-85%) and found the velocity to be marginally lower but yet stable (0.12-0.14 m/s). Loturco et al. (2017) investigated the difference in minimal velocity threshold in the bench press conducted with a Smith machine and with free-weights. The authors hypothesised a difference, due to the contribution of the stretch-shortening cycle in the free-weight exercise which could result in a higher force production. However, the difference was not significantly higher, 0.15 m/s for the Smith machine and 0.18 m/s for the free-weight bench press.

Sánchez-Medina et al. (2014) compared the minimal velocity threshold in two antagonistic upper body exercises, the bench press and the bench pull. The results showed that the minimal velocity threshold for the bench press was 0.17 m/s (as in previous studies) and 0.52 m/s for the bench pull. The authors suggest, based on these results, that the minimal velocity threshold is exercise specific.

These results suggest that the minimal velocity threshold remains the same, independently of relative load being lifted, exercise selection or if its performed using free-weight or a Smith machine. Nevertheless, the minimal velocity threshold is highly exercise specific.

Table 4. The mean velocity reached before failure. 1RM= 1 repetition maximum.

Study	Exercise	Equipment	Load (% 1RM)	Velocity (m/s) mean ± SD
González-Badillo & Sánchez-Medina (2010)	Bench press	Smith machine	100%	0.19 ± 0.04
Sánchez-Medina et al. (2014)	Bench press	Smith machine	100%	0.17 ± 0.04
	Prone bench pull		100%	0.52 ± 0.06
Loturco et al. (2016)	Half Squat	Smith machine	100%	0.31 ± 0.02
Loturco et al. (2017)	Bench press	Smith machine	100%	0.15 ± 0.03
		Free weight	100%	0.18 ± 0.02
Izquierdo et al. (2006)	Bench press	Smith machine	60%	0.17 ± 0.06
			65%	0.18 ± 0.05
			70%	0.18 ± 0.05
			75%	0.17 ± 0.04
	Parallel Squat		60%	0.33 ± 0.07
			65%	0.31 ± 0.06
			70%	0.32 ± 0.70
			75%	0.31 ± 0.05
			50%	0.14 ± 0.03
			55%	0.14 ± 0.04
			60%	0.13 ± 0.02
			65%	0.14 ± 0.04
González-Badillo et al. 2017	Bench press		70%	0.13 ± 0.03
			75%	0.13 ± 0.02
			80%	0.12 ± 0.02
			85%	0.14 ± 0.02

Validity of different linear position transducers

Banyard et al. (2017) was the only study that used a combination of both linear position transducers and a force plate. Their objective was to validate the Push band and GymAware for assessing mean velocity, peak velocity, mean force, peak force, mean power and peak power during a full-depth free-weight back squat on 10 strength-trained men. This study was conducted on 6 different intensities (% 1RM) to better reflect an athletes varying intensities during the various phases of a periodized training plan. The results revealed that the GymAware was considered valid in the assessment of all criterion variables, whereas the Push band was only valid to estimate peak force. However, the data regarding the correlation between the two linear position transducers and the reference system is not presented in absolute values by the author, which makes it difficult analyse and draw any own conclusion based on the results.

The T-Force has both been validated against gold standard and functioned as a gold standard itself. To the authors knowledge, the T-Force was first used to validate linear position transducers in 2015 by Garnacho-Castaño, López-Lastra and Maté-Muñoz, but was not validated against a force plate until the year later by García-Ramos et al. (2016). However, when later validated by García-Ramos et al. (2016), it was found to significantly correlate in mean velocity, peak velocity, mean force, peak force, mean power and peak power at four load ranging from 25-100% of 1RM.

Lorenzetti, Lamparter & Lüthy 2017 validated the T-Force, Tendo Power, GymAware Myotest Pro with a 3D motion capture system. All linear position transducers were found to significantly correlate with the 3D system when measuring peak- and mean velocity. This could be explained by the fact that the linear position transducer measures displacement over time, just like the 3D system.

Altogether, the GymAware and the T-Force were found to be valid and reliable for both kinetic and kinematic assessments.

Table 5. Validation of different linear position transducers. MV= mean velocity, PV= peak velocity, MF= mean force, PF= peak force, MP= mean power, PP= peak power, R= correlation of coefficient, ICC= intraclass correlation. Significant correlation: * $P = <0.01$, ** $P = <0.001$.

Author	Device	Measurement	Exercise	Reference Instrument	R	ICC
Balsobre-Fernández et al. 2016	Push band	PV, MV	Back Squat	T-Force	PV = 0.91**, MV = 0.86**	MV = 0.91, PV = 0.94
O'Donnell et al. 2018	GymAware	Jump Height (cm)	Counter-movement jump	Force Plate	0.90*	0.70
Garnacho-Castano, López-Lasra & Maté-Muñoz 2015	Tendo Power	MV, PV, MP, PP	Bench press Back Squat	T-Force		MV = 0.99, PV = 0.98, MP = 0.97, PP = 0.9 MV = 0.96, PV = 0.96, MP = 0.97, PP = 0.85
Banyard et al. 2017	Push band GymAware	MV, PV, MF, PF, MP, PP	Back squat	4 LPTs and Force plate		
	GymAware				MF = 0.98, MV = 0.91, MP = 0.89	MF = 0.96, MV = 0.86, MP = 0.89
Giroux et al. 2015	Myotest Pro (Myotest)	MV, MF, MP	Squat Jumps	Force Plate	MF = 0.98, MV = 0.87, MP = 0.89	MF = 0.97, MV = 0.84, MP = 0.89
	Optolump Next (Microgate)				MF = 0.98, MV = 0.88, MP = 0.87	MF = 0.99, MV = 0.97, MP = 0.97
García-Ramos et al. 2016	T-Force (Ergotech)	MV, PV, MF, PF, MP, PP	Loaded jumps	Force Plate	$r = 0.83-0.96^{**}$	0.86-0.97
	T-Force (Ergotech)					
	Traditional squat: MV = 0.98*, PV = 0.93*, t PV 0.98*. Ballistic squat: MV = 0.72*, PV = 0.81*, t PV = 0.65*					
	Traditional squat: MV = 0.96*, PV = 0.93*, t PV 0.98*. Ballistic squat: MV = 0.77*, PV = 0.86*, t PV = 0.60*					
Lorenzetti, Lamparter & Lüthy 2017	Tendo Power					
	GymAware	MV, PV, Time to Peak Velocity (t PV)	Back Squat + Ballistic Squat	3D motion capture system	Traditional squat: MV = 0.95*, PV = 0.95*, t PV = 0.99*. Ballistic squat: MV = 0.78*, PV = 0.85*, t PV = 0.70*	
	Myotest Pro (Myotest)				Ballistic squat: MV = 0.61*, PV = 0.55*, t PV = 0.70*	

Neuromuscular effects of velocity-based training

All studies presented below used a squat-only protocol during the training intervention, the variation came from manipulating the intensity and volume.

Pareja-Blanco et al. (2017ab) utilized a protocol where the subject stopped the exercise when a given velocity loss was reached at different relative intensities (e.g. VL20, VL40, VL15 and VL30). Pareja-Blanco et al. (2014) did however use maximal intended velocity (MaxV) and half-maximal velocity (HalfV) in their intervention. In all 3 studies, the velocities at certain relative intensities was predetermined before the start of the intervention.

Pareja-Blanco et al. (2017a) got significant results for both VL20 (pre 106.5 ± 12.2 , post 125.2 ± 12.3 , $p < 0.001$) and VL40 (pre 104.5 ± 15.1 , post 118.6 ± 20.4 , $p < 0.001$) after 16 sessions in 8 weeks. Similar results were found by Pareja-Blanco et al. (2014) when using max velocity (pre 89.2 ± 15.0 , post 105.2 ± 18.0 , $p < 0.001$) and half velocity (pre 94.8 ± 17.0 , post 104.0 ± 17.0 , $p < 0.01$) after a more intensive training regime with 18 sessions in 6 weeks. The same result was also found by Pareja-Blanco et al. (2017b), but with VL15 (pre 101.3 ± 18.8 , post 110.3 ± 14.3 , $p < 0.01$) after 18 sessions in 6 weeks. However, no significant difference was found for the VL30.

The mean propulsive velocity was significantly higher in the squat after the intervention for the VL20 (pre 0.95 ± 0.06 , post 1.06 ± 0.06 , $p < 0.001$), VL40 (pre 0.95 ± 0.06 , post 1.01 ± 0.09 , $p = 0.03$), MaxV (pre 1.03 ± 0.05 , post 1.18 ± 0.11 , $p < 0.001$) and HalfV (pre 1.06 ± 0.08 , post 1.14 ± 0.10 , $p < 0.01$). No significant change in mean propulsive velocity was found in the study from Pareja-Blanco et al. (2017b), even if the VL15 had significantly improved their 1RM.

The greatest increase in countermovement jump came in VL20 (pre 40.5 ± 6.0 , post 44.2 ± 6.0 , $p < 0.001$), VL15 (pre 33.7 ± 3.6 , post 35.5 ± 5.1 , $p < 0.05$) and MaxV (pre 36.6 ± 4.5 , post 39.9 ± 5.8 , $p < 0.001$). Given these results, it could be argued that to improve CMJ the subjects need to train with high velocities and in a non-fatigue state.

MaxV was found to be beneficial for improving 10 m (pre 1.80 ± 0.07 , post 1.75 ± 0.06 , $p < 0.01$) and 20 m (pre 3.09 ± 0.10 , post 3.04 ± 0.10 , $p < 0.05$). The HalfV was also found to significantly improve 20 m sprint time (pre 3.07 ± 0.13 , post 3.02 ± 0.12 , $p < 0.05$).

To summarize, using maximal intended velocity or stopping when a given velocity loss ($<40\%$) is reached in the squat can significantly improve the 1 repetition maximum, mean propulsive velocity during a set of squat and vertical jump performance. Conflicting results exist regarding the protocols benefits on sprint performance.

Table 6. Neuromuscular effects of velocity-based training. VL= velocity loss, MaxV= maximal intended velocity, HalfV= half of maximal intended velocity, CMJ= countermovement jump, 1RM= 1 repetition maximum, MPV= mean propulsive velocity, T10= 10 m sprint time, T20= 20 m sprint time, T30= 30 m sprint time. Significant difference from pre- to post-training: * $P = <0.05$, ** $P = <0.01$, *** $P = <0.001$.

Name	Subjects	Duration	Exercise	Protocol	CMJ (cm)		1RM (kg)		MPV in Squat (m/s)		T10		T20		T30	
					Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Pareja-Blanco et al. 2017a	22 males	8 weeks (16 sessions)	Squat	VL20 VL40	41 ± 6 41 ± 4	44 ± 6* 42 ± 6.7	107 ± 12 105 ± 15	125 ± 12** 119 ± 20**	1.0 ± 0.1 1.0 ± 0.1	1.1 ± 0.1** 1.0 ± 0.1*			3.0 ± 0.1 3.0 ± 0.1	3.0 ± 0.1 3.0 ± 0.1		
Pareja-Blanco et al. 2014	21 males	6 weeks (18 sessions)	Squat	MaxV HalfV	37 ± 5 41 ± 6	40 ± 6** 42 ± 7	89 ± 16 95 ± 17	105 ± 18** 104 ± 17*	1.0 ± 0.1 1.1 ± 0.1	1.2 ± 0.1** 1.1 ± 0.1*	1.8 ± 0.1 1.8 ± 0.1	1.8 ± 0.1* 1.8 ± 0.1	3.1 ± 0.1 3.1 ± 0.1	3.0 ± 0.1* 3.0 ± 0.1*		
Pareja-Blanco et al. 2017b	16 males	6 weeks (18 sessions)	Squat	VL15 VL30	34 ± 4 34 ± 4	36 ± 5* 34 ± 3	101 ± 19 100 ± 20	110 ± 14* 107 ± 29	1.2 ± 0.1 1.2 ± 0.1	1.2 ± 0.1 1.2 ± 0.1					4.3 ± 0.2 4.3 ± 0.14	4.3 ± 0.2 4.3 ± 0.10

Discussion

Result discussion

The main finding of this systematic review suggest that there exists a strong correlation between the 1 repetition maximum and velocity. Performing the squat with maximal velocity or stopping when a small (<40%) velocity loss is reached, can significantly improve neuromuscular performance after six to eight weeks of training.

When assessing the 1 repetition maximum using velocity as a measurement of intensity, 3-6 different loads ranging from 20-95% of the known 1RM with approximately 3 minutes of rest in between sets is preferred. This is in line with the work from both Jovanović and Flanagan (2014) and Jidovtseff et al. (2011) whom recommended the use of 4-6 different loads ranging from 30-85% of the 1RM and with 3 minutes of passive recovery. It should be added, that using heavier loads increases the validity of the prediction compared to lighter loads due to the contribution of the stretch-shortening cycle (Turner & Jeffreys 2010).

Although Banyard and colleagues (2017) found a strong correlation between load and velocity ($r = 0.78-0.93$, $r^2 = 61-87\%$), all predicted 1RMs were significantly different from the actual 1 repetition maximum ($p \leq 0.05$). Similar results have been found by Ruf, Chéry & Taylor (2018) who found that the predictions tended to overestimate the actual 1RM with approximately 5-10 kg and high individual differences. These results could be related to the usage of free weight exercises. It has been suggested that use of a counter-balance mechanism (as in free weight) results in significantly lower concentric velocities and accelerations and may therefore underestimate the capability (Buddhadev et al. 2012; Vingren, Buddhadev & Hill 2011) Unfortunately, Loturco et al. (2017) did not analysed the prediction of the 1 repetition maximum when comparing the bench press performed with free weight with the Smith machine. More research is needed in this area before any clear conclusion can be drawn.

The weakest correlation ($r = 0.78$, $r^2 = 61\%$) between load and velocity was found by Banyard, Kazunori & Nosaka (2017) when attempting to predict the 1RM from a protocol using loads up to 60% of 1RM. The authors discuss that this may be attributed to the higher contribution of the stretch-shortening cycle, which is caused by the recoiling action of the move-

ment that may increase the concentric velocity and therefore influence the prediction of the 1RM (Turner & Jeffreys 2010).

As the minimal velocity threshold was found to be stable regardless of the weight being lifted or exercise selection, it is possible for coaches to stop athletes before reaching failure. This is of great value as reaching failure during strength training has been associated with a higher risk of injuries (Jovanović & Flanagan 2014). Depending on the exercise, the minimal velocity threshold appeared to be different, suggesting that it is highly exercise specific. A good example of the specificity of the minimal velocity threshold would be the study from Sánchez-Medina et al. (2014) where two antagonistic muscles groups were compared to each other (bench press versus prone bench pull). It is discussed by the authors that this may be related to the different skeletal muscle architecture and mechanical levers. Additionally, different sticking regions, which is the point where the lengths and moment arms of the muscles involved are such that their capacity to exert force is reduced, may have contributed to the results. In conclusion, minimal velocity threshold seems to be stable but highly exercise specific.

Recent research has evaluated whether the mean velocity of an object can be used to determine the exact number of repetition left in reserve in a training set (García-Ramos et al. 2018; González-Badillo et al. 2017). However, conflicting results exist between these two studies regarding its reliability. The varying results in García-Ramos et al. (2018) and González-Badillo (2017) may be attributed to the different equations utilized to estimate the repetitions left in reserve. More research is required before any conclusion can be drawn.

During several investigations, both mean velocity and mean propulsive velocity was used when determining the velocity of an object. Sanchez-Medina, Perez & Gonzalez-Badillo (2010) have highlighted this issue and concluded that mean velocity tends to overestimate the velocity of the barbell. The need to measure the propulsive phase becomes even more important during both lighter and medium loads due to the fact that higher velocities needs longer braking phases. Evidence of this has been presented by Sánchez-Medina et al. (2014), whom measured the contribution of both the braking- and propulsive-phase during load ranging from 30-100% of 1RM in bench press and prone bench pull. It was concluded that, at higher loads (85-100% of 1RM) the magnitude of the braking phase was minimal whereas the propulsion phase was high and vice versa for light loads (e.g. 30% of 1RM).

On the basis of the above mentioned, one can question the validity of the previous conducted investigations that have used mean velocity as a measurement instead of mean propulsive. For example, a person with a higher 1RM will lift the same absolute loads at faster velocities and will thus experience a longer braking phase that negatively influences the computation of the mean concentric value since force, and hence power output, is negative during the braking phase (Sanchez-Medina, Perez & Gonzalez-Badillo 2010).

The results from the interventions suggested that several neuromuscular adaptations can be obtained when using velocity as a measurement of intensity. Previously published literature, indicates not reaching neuromuscular failure has positive effects on countermovement jump performance (Morán-Navarro et al. 2017; González et al. 2016; Sánchez-Medina & González-Badillo 2011). The three experimental studies used in this review support the notion that using high velocities or stopping when a loss of velocity of <40% is reached, significantly increases countermovement jump performance. The mean concentric velocity of the squat was significantly improved when the subjects stopped the exercise at a loss of velocity corresponding to 20% and 40% was reached. The same results were found by Pareja-Blanco et al. (2014) when performing the squat at with both maximal velocity and half of the maximal velocity. Surprisingly, stopping at a velocity loss of 15 and 30% did not significantly improve their mean velocity. There was no improvement in the 1 repetition maximum when a velocity loss of 30% was reached, however, significant improvements were found in all other groups which suggest that not reaching failure and monitoring the velocity could result in neuromuscular improvements.

Using maximal velocity (MaxV) during the squat significantly improved 10- and 20-meter sprint time and training with half of the maximal velocity (HalfV) improved 20-meter sprint time. These results are conflictingly, as previous research by Wisloff et al. (2004) has shown a strong correlation between the 1RM and 10-meters sprint time ($r = 0.94$, $p < 0.001$) and 30-meters sprint time ($r = 0.71$, $p < 0.01$). Additionally, sprint time on both 10- and 30-meter correlated with vertical jump performance ($r = 0.72$, $p < 0.001$; $r = 0.60$, $p < 0.01$). Furthermore, Styles, Matthews and Comfort (2016) found that improving the back squat 1RM had significant effects on 5-, 10- and 20-meter sprint time ($p \leq 0.001$). However, the results found in this study are not in agreement with the aforementioned, therefore, more research on this field is needed before any conclusion can be drawn.

This review did not examine the metabolic effect of different intervention using velocity monitoring. However, a cross-sectional study by Sánchez-Medina and González-Badillo (2011) found that lactate and ammonia levels were elevated as the number of repetitions increased. Both González-Badillo et al. (2016) and Morán-Navarro et al. (2017) examined the relationship between velocity loss and changes in testosterone levels and concluded that training to muscle failure does not provide a superior stimulus to increase blood testosterone levels over non-failure sets to speed up recovery. Moreover, González-Badillo (2011) found elevated levels of creatine kinase which is associated with skeletal muscle damage (McGuigan & Cormack 2011, p. 309). Long term metabolic effects following a velocity specific intervention has not yet been thoroughly examined, and more research on this area is wanted.

Several authors have suggested that the best method to validate a linear position transducer is with a use of a force plate or, preferably, a combination of a force plate and a linear position transducer (Cormie et al. 2007; Hansen et al. 2011; Nimphius 2017, p. 26). Using only kinematic measurements (e.g. a linear position transducer) may overestimate power output (Nimphius 2017, p. 26). Based on these validation criteria for a linear position transducer, the only study utilizing a combined method of the two was Banyard et al. (2017) when validating the GymAware and the Push and found the GymAware to be valid for all measurements (table 5). Furthermore, the GymAware has been found valid and reliable for both kinetic and kinematic measurements when compared to force plates and 3D motion system, thus making it a justifiable measuring tool.

To the author's knowledge, there is no study that have thoroughly validated the T-Force. However, Sánchez-Medina and González-Badillo (2011) noted that the T-Force was found to be both valid and reliable for displacement measurement when compared to a high-precision digital height gauge that had been calibrated by the Spanish national institute of aerospace technology. When compared against 18 other T-Force units, displacement was accurate to ± 0.5 mm. Additionally, when performing 30 repetitions with two devices simultaneously an ICC of 1.00 were obtained for mean propulsive velocity, and an ICC of 1.00 for peak velocity. This makes the T-force valid and reliable for assessing velocity, however no reference is presented in the study where the results can be analysed. Lorenzetti, Lamparter & Lüthy (2017) validated the T-Force against a 3D motion system and found a significant correlation for mean velocity, peak velocity and time to peak velocity in back squat ($r= 0.93-0.98$) and

ballistic squat ($r = 0.65-0.81$) suggesting that the T-Force may be a reliable tool for kinematic assessments.

The wearable Push device was found to be valid and reliable for assessing mean velocity and peak velocity when using the T-Force as a reference instrument. Both peak velocity and mean velocity had a significant correlation ($r = 0.91$ and 0.86) with the T-Force (Balsalobre-Fernández et al. 2016). This is of course in line with the results obtained from Sánchez-Medina and González-Badillo (2011) who found the T-Force to be justifiable for assessing velocity. However, when the Push device was tested for peak and mean velocity using four linear position transducers and a force plate it was not accurate enough (Banyard et al. 2017). One possible reason for this may be due to the Smith machine used in the study from Balsalobre-Fernández et al. (2016) compared to the free-weight in Banyard et al. (2017). Given that the Push was only found to be valid for mean velocity at loads $<60\%$ of 1RM and peak velocity at 20% of 1RM, the accuracy of the Push may be depending on the relative intensity, which may be a concern for athletes whom train at a variety of loads throughout the training year (Haff and Nimphius 2012).

There is to this date, no study that has investigated the velocity during unilateral movements. Bilateral leg exercises are performed with a two leg base of support, however, athletic skills such as sprinting, jumping and changing direction are performed either with one leg at a time or with weight transferred to one leg at a time. Because of this, it could be speculated that unilateral exercises are preferable for improving these skills compared to bilateral exercises given the principle of training specificity (Speirs et al. 2016; Nijem & Galpin 2014). However, there is some controversy to this matter as a decreased stability could be viewed as a limitation in resistance training because of the difficulty to prescribe heavy loads, which could lead to lower expressions of force (Behm and Anderson 2006). It should be added that unilateral plyometric training has been proven to improve greater unilateral jump height/length (Ramírez-Campillo 2015; Makaruk et al. 2011). The authors discuss the importance of training specificity, as unilateral exercises induced greater enhancements in unilateral dominant exercises, whereas bilateral exercises induced greater bilateral improvements. This can also be described as the “transfer effect” or “transferability” of a given exercise/programme (Verkhoshansky & Siff 2009, p. 241). Given the above mentioned possible benefits of using unilateral exercises for athletic performance, no study has yet combined

unilateral exercises and a linear position transducer. The author of this review suggest that more research in needed is this area before any conclusion is drawn.

Methodological Quality

Among the inclusion criteria, no considerations were taken to the subjects' age and sex. Subjects had to have at least 6 month of resistance training experience, which could have influenced the results, especially during the interventions, as more novice subjects will gain faster neuromuscular adaption then more experienced subjects (Kenney, Wilmore & Costill 2015, p. 245). Furthermore, if the resistance training experience was not specified, but the subjects were athletes, it was considered valid enough which could have led to a source of error. Even if no date restrictions were used, 17 out of 19 studies were between 2014-2018. Since the latest review regarding velocity-based training was published in 2014 by Jovanović and Flanagan, it makes this study up-to-date.

The results that are presented in the studies are very variable, this has to do with the different statistical analyses and presentation of results (e.g. absolute values, relative values, graphs and/or figures) by the authors. The varying forms of which some results are presented, may have contributed to a source of error due to the lack of consistency.

A quality assessment using the Oxford Centre of Evidence Based Medicine levels of Evidence 2011 was conducted. Three studies were considered to have a high methodological quality whereas 16 had moderate methodological quality. This may be explained by the fact that the three studies with a high methodological quality had a randomized controlled study design, compared to the others (16) that had a cross-sectional study design. Furthermore, the studies included in this review were not supervised by an independent part, which could have resulted in a systematic bias.

As determining the minimal velocity threshold of an exercise was not the focal point of all of the studies, the results in question were rather the bi-product of other objectives. The results may have been otherwise with a different aim. Furthermore, only three studies investigated the neuromuscular effects following an intervention, the results should therefore be treated with caution.

Future research

Future research should investigate the prediction of 1RM using a linear position transducer with free-weight exercises, as free-weight equipment is used more frequently by athletes and is relative inexpensive compared to a Smith machine. More commercial linear position transducer needs to be validated against, preferably, a force plate and a linear position transducer. At the moment, many linear position transducers are able to predict velocity, but fail to accurately measure force and power output.

As using high velocity or stopping when a given loss of velocity is reached had great neuromuscular benefits on squat and vertical jump performance, future research should investigate the metabolic response using the same protocols. Due to the conflicting results regarding improvements in the back squat and sprint time, there is a need to further examine this issue. Performing exercises with a horizontal force vector instead of a vertical may present different adaptations. Furthermore, no study has yet investigated the effects of velocity-based training on unilateral exercises, which may be desirable to increase the specificity for athletes.

Conclusion

The main finding of this systematic review suggest that there exists a strong correlation between the 1 repetition maximum and velocity and that the minimal velocity threshold is stable across different relative intensities and can be used to avoid reaching failure. Performing the squat at either maximal velocity, or stopping at a velocity loss of 20 and 40% can significantly improved the 1 repetition maximum, mean propulsive velocity during a set of squats and vertical jump performance. The GymAware is the only commercial linear position transducer that has been validated against both a force plate and linear position transducers and found to be valid for both kinetic and kinematic measurements. Although the T-Force has not been tested as thoroughly as the GymAware, it has been deemed valid and reliable when tested with both force plates and 3D optical system separately.

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Appendices

Appendix 1 – Literature research details

Keywords, database, number of hits and date used in the literature research

Keyword	Database	Number of Hits	Date
Linear position transducer	PubMed	408	12-04-2018
Velocity based training	PubMed	1 260	12-04-2018
Power development	PubMed	29 496	12-04-2018
Movement velocity	PubMed	23 843	14-04-2018
Force-velocity	PubMed	1 519	14-04-2018
1 repetition maximum	PubMed	3 781	14-04-2018
Linear position transducer	SportDiscus	404	12-04-2018
Velocity based training	SportDiscus	1 588	12-04-2018
Power development	SportDiscus	186 314	12-04-2018
Movement velocity	SportDiscus	23 502	14-04-2018
Force-velocity	SportDiscus	1 424	14-04-2018
1 repetition maximum	SportDiscus	5 103	14-04-2018
Velocity based training	ReserachGate	Not presented	15-04-2018
Linear position transducer	ReserachGate	Not presented	15-04-2018

Relevant combination of keywords

Keyword	Database	Number of Hits	Date
force-velocity + 1 repetition maximum	SportDiscus	86	22-04-2018
force-velocity + 1 repetition maximum	PudMed	37	21-04-2018
linear position transducer + 1 repetition maximum	SportDiscus	39	22-04-2018
linear posititon transducer + power	SportDiscus	121	22-04-2018
movement velocity + linear position transducer	SportDiscus	35	22-04-2018
movement velocity + 1 repetition maximum	SportDiscus	262	22-04-2018

The number of articles found in databases, books, reference lists and thru direct contact with the authors that were included in this study.

Source	Number of articles
Databases	16
Books	0
Reference lists	3
Direct contact with authors	0

Appendix 2 – List of included articles

Year	Author	Title
2006	Izquierdo et al.	Effect of Loading on Unintentional Lifting Velocity Declines During Single Sets of Repetitions to Failure During Upper and Lower Extremity Muscle Actions
2010	González-Badillo & Sánchez-Medina	Movement velocity as a measure of loading intensity in resistance training.
2014	Sanchez-Medina et al.	Velocity- and Power-Load Relationships of the Bench Pull vs. Bench Press Exercises.
2014	Pareja-Blanco et al.	Effect of movement velocity during resistance training on neuromuscular performance.
2015	Giroux et al.	What is the best method for assessing lower limb force-velocity relationship?
2015	Garanchó-Castaño, López-Lastra & Maté-Muñoz	Reliability and Validity Assessment of a Linear Position Transducer.
2016	Balsalobre-Fernández et al.	Validity and reliability of the PUSH wearable device to measure movement velocity during the back squat exercise.
2016	Loturco et al.	Using Bar Velocity to Predict Maximum Dynamic Strength in the Half-Squat Exercise.
2016	Conceição et al.	Movement velocity as a measure of exercise intensity in three lower limb exercises.
2016	García-Ramos et al.	Comparison of the force-, velocity-, and power-time curves recorded with a force plate and a linear velocity transducer.

2017	Loturco et al.	Predicting the maximum dynamic strength in bench press: the high precision of the bar velocity approach
2017	Banyard, Nosaka & Haff	Reliability and Validity of the Load- Velocity Relationship to Predict the 1RM Back Squat
2017	González-Badillo et al.	Velocity Loss as a Variable for Monitoring Resistance Training.
2017	O'Donnell et al.	The validity and reliability of the GymAware linear position transducer for measuring counter-movement jump performance in female athletes
2017	Banyard et al.	Validity of Various Methods for Determining Velocity, Force, and Power in the Back Squat
2017	Lorenzetti, Lamparter & Lüthy	Validity and reliability of simple measurement device to assess the velocity of the barbell during squats
2017	Pareja-Blanco et al.	Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations
2017	Pareja-Blanco et al.	Effects of Velocity Loss During Resistance Training on Performance in Professional Soccer Players
2018	Ruf, Chéry & Taylor	Validity and reliability of the load- velocity relationship to predict the 1RM in deadlift.

